Graph Meta Transfer Network for Heterogeneous Few-Shot Hyperspectral Image Classification

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***Abstract*— Since obtaining labeled hyperspectral images (HSIs) is difficult and time-consuming, the shortage of training samples has always been a challenge for HSI classification. In practical applications, only a few labeled samples are available in the task domain (target domain), while sufficient labeled samples are available in another domain (source domain). At the same time, these two domains are heterogeneous and contain different categories. This scenario makes it difficult to effectively transfer knowledge from the source domain to the target domain. To address this challenge, we propose a novel heterogeneous few-shot learning (FSL) method, namely graph meta transfer network (GMTN). Specifically, the graph sample and aggregate network (GraphSAGE) and meta-learning, which are both inductive learning, are integrated into a unified framework. In this way, the aggregation function is generalized from abundant few-shot tasks for feature extraction on the source and target domains. The spatial importance strategy (SIS) is designed to guide the feature propagation and alleviate the information interference caused by different categories. The neighborhood receptive field spectral attention (RFSA) mechanism is proposed to model the importance of spectral band using the information of the neighborhood pixels, which enables GMTN to pay more attention to bands with discriminative features in both domains. In addition, the node spatial information reset method is proposed to augment samples based on the spatial position relationship of nodes. Furthermore, to alleviate the domain shift in heterogeneous scenarios, the conditional domain adversarial strategy is used to achieve effective meta-knowledge transfer. Experiments show that GMTN outperforms the compared state-of-the-art methods.**

***Index Terms*— Attention mechanism, few-shot learning (FSL), heterogeneous, hyperspectral image (HSI) classification, meta-learning, transfer learning.**

I. Introduction

HYPERSPECTRAL image (HSI) classification aims to utilize spectral and spatial information to judge the category of each pixel, which helps human beings better understand their living environment [1], [2]. Machine learning can obtain rules by automatically analyzing data and then using the obtained rules to predict unseen data. In view of the above ability, machine learning has been successfully applied to HSI classification and has achieved remarkable research results [3].

The traditional machine learning methods used in HSI classification include support vector machine (SVM) [4], k-nearest neighbor (KNN) [5], and so on. These methods cannot obtain high accuracies due to their limited feature extraction abilities.

In recent years, deep learning has been widely used in HSI classification tasks [6], [7]. Compared with traditional machine learning methods, deep learning can learn meaningful factors from data, thereby helping models achieve higher classification accuracies [8]. Many recognized backbone networks have been widely used for HSI classification, including: autoencoders [9], convolutional neural networks (CNNs) [10], generative adversarial network (GAN) [11], and graph neural network (GNN) [12]. Chen et al. [9] proposed a deep learning-based HSI classification model by combining the autoencoder and principal component analysis (PCA), which can take advantage of both the spatial and spectral information. Yu et al. [10] proposed a dual-channel convolution network (DCCN) with global feature learning module and multiscale feature fusion module, which can fully exploit the global and multiscale local information of HSI. Zhu et al. [11] proposed a GAN-based HSI classification method where the adversarial training is performed through the zero-sum game between discriminator and generator. The generator is used to generate realistic augmented samples, and then the real samples and augmented samples are used for HSI classification. Ding et al. [12] captured rich spatial information by adaptively searching for the global high-order neighborhoods and exploiting the graph topological consistency. The GNN can effectively extract, aggregate, and transform the neighborhood information of each graph node by modeling the graph vertices. Given their powerful spatial–spectral feature learning capability, GNNs are well suited for HSI classification tasks [13].

To date, the quantity and quality of data are key factors for machine learning to make an accurate prediction. However, labeling hyperspectral data is time consuming and labor consuming; therefore, it is very difficult to build a large-scale and ideal labeled training set [14]. How to fulfill HSI classification

Manuscript received 6 November 2022; revised 13 December 2022; accepted 26 December 2022. Date of publication 2 January 2023; date of current version 11 January 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 61976215 and Grant 62176259, in part by the Key Research and Development Program of Jiangsu Province under Grant BE2022095, in part by the Natural Science Foundation of Jiangsu Province under Grant BK20221116, and in part by the Excellent Post Doctorate Program of Jiangsu Province under Grant 2022ZB530. *(Corresponding author: Yuhu Cheng.)*

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Digital Object Identifier 10.1109/TGRS.2022.3233591

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at a low labeling cost is a hot topic. Active learning, semisupervised learning, and data augmentation methods have been proposed to solve this problem. However, none of the above methods can handle the situation when the training and testing sets are heterogeneous. Heterogeneous data are characterized by different sets of features, that is, their data dimensions and distributions are different [15]. In reality, due to the differences in shooting sensors, shooting areas, and other factors, the HSIs collected are often heterogeneous and contain different categories. This scenario makes it difficult to transfer knowledge effectively. Therefore, how to use heterogeneous HSIs with different categories to assist in few-shot HSI classification is a meaningful research topic.

Transfer learning can utilize labeled data in a certain domain (source domain) to accomplish tasks in a similar but different domain (target domain), regardless of whether the source and target domains are homogeneous or heterogeneous [16], [17], [18]. Common transfer methods applied to HSI classification include: deep adaptation network (DAN) [19], domain adversarial network (DANN) [20], broad domain adaptation network, and graph domain adaptation network. Wang et al. [19] designed a deep domain adaptation model to adapt the two-domain distributions while maintaining the manifold structure of original data. Liu et al. [20] proposed a category-level distribution adaptive model, which introduce a probability-prediction-based maximum mean discrepancy (PMMD) method to the adversarial adaptation network, and completed the classification with low labeling cost. Wang et al. [21] aligned two domains by comprehensively reducing their statistical differences. Kong et al. [22] proposed a graph DANN, which reduced the distribution difference between domains in an adversarial manner while making full use of the spatial–spectral information. Makkar et al. [23] applied the domain adversarial ideal to remote sensing image processing, and achieved high accuracies on HSI and multispectral image classification tasks. Elshamli et al. [24] applied the domain adaptation representation learning directly to remote sensing image processing. Kalita et al. [25] proposed a deep domain adaptation method, which used active learning to select target-domain samples with abundant information to augment the training set. Transfer learning can solve the heterogeneous situation of two domains. However, it is difficult to solve the problem of classification when two-domain data contain different categories.

Meta-learning is a machine learning method of "learning how to learn." It uses few labeled samples to learn the commonness and specificity between tasks and exploits the knowledge learned in previous tasks to quickly adapt and complete new tasks [26], [27], [28]. In view of the above advantages, a small number of researchers have successfully applied meta-learning to few-shot HSI classification. Liu et al. [29] first enabled the network to extract spatial–spectral features with high interclass dissimilarity through meta-training, then mapped them to subspaces, and finally completed classification by measuring the distance between the samples from the support and query sets. Rao et al. [30] proposed a spatial-spectral relationship network, which generalized the metric function between different samples, and applied it to the few-shot HSI classification. Bai et al. [31] proposed a few-shot classification model based on subspace and feature transformation, combined with a feature transformation strategy to enhance feature diversity, and learned a classifier that can quickly adapt to unseen samples through meta-strategy. Li et al. [32] proposed a novel deep cross-domain few-shot method, which combined the conditional adversarial domain adaptation strategy with meta-learning, and achieved the heterogeneous few-shot HSI classification.

In summary, although meta-learning can achieve few-shot HSI classification to a certain extent, the following problems still exist: 1) it is difficult for meta-learning based classification networks to acquire knowledge with spatial information and migrate it to new tasks and data and 2) in the heterogeneous few-shot scenario, the model learned from meta-tasks is difficult to accurately extract the discriminative features of new heterogeneous data. To address these issues, based on meta-learning and transfer learning, we proposed a heterogeneous few-shot HSI classification method named the graph meta transfer network (GMTN). The contributions include the following.

1) We integrated graph learning, meta-learning, and transfer learning into a unified framework, which can effectively induce transferable meta-knowledge containing spatial–spectral information from few-shot tasks and apply it to the classification of heterogeneous HSIs.
2) We proposed a neighborhood spatial information reset (NSIR) method that can not only endow the augmented samples with rich spatial information but also make the augmented samples aggregate the diverse adjacent spectral information as much as possible.
3) We proposed a neighborhood receptive field spectral attention (RFSA) mechanism. By reducing the interference of spectral band importance learning caused by different-label neighborhood pixels, more effective spectral band importance learning can be realized, thereby helping the GMTN extract discriminative features from the perspective of spectral information.
4) We proposed a spatial importance strategy (SIS). Based on the spectral feature similarity and distance constraint, the spatial importance weights of neighborhood pixels were given to make the feature aggregation pay more attention to the useful information, thus improving the ability of GMTN to extract the discriminative spatial-spectral features.

II. Proposed GMTN

A. Network Overview

The GMTN is mainly composed of six components, including PCA, NSIR, RFSA, SIS, graph sample and aggregate network (GraphSAGE), and conditional domain adaptation. First, PCA is used to unify the spectral dimensions of HSIs from the source and target domains. Second, NSIR is used to augment the target-domain data. Third, the meta-training set is constructed for few-shot learning (FSL). During the FSL process, RFSA and SIS are used to model importance of spectral band and weight neighborhood pixel, respectively, to help

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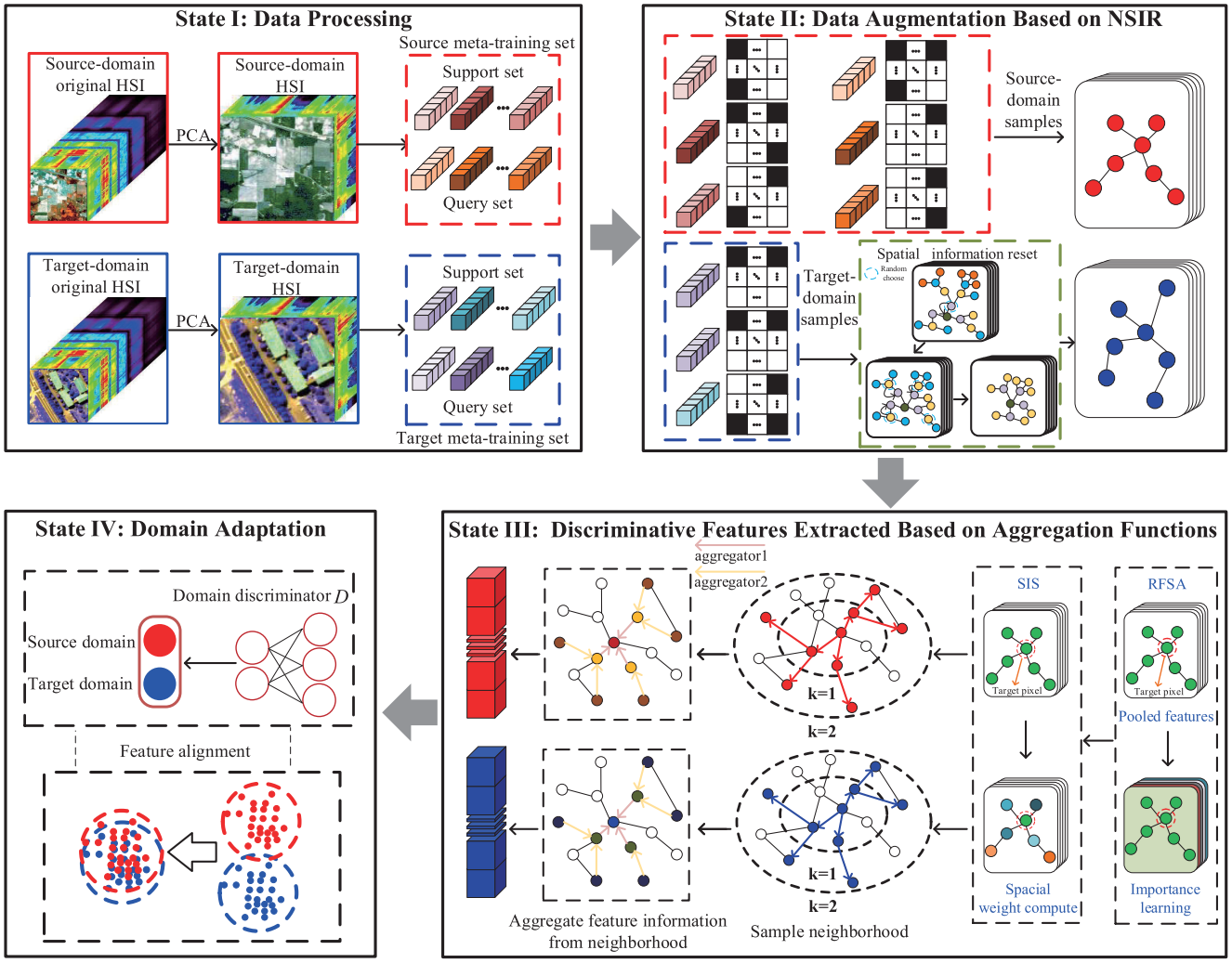

Fig. 1. Training phase of GMTN.

GraphSAGE extract more discriminative spatial–spectral features. Meanwhile, the feature distributions of both domains are aligned through the conditional domain adaptation strategy. Finally, the heterogeneous few-shot HSI classification is fulfilled.

B. Method Flow

The training phase of the GMTN is shown in Fig. 1, which is divided into four key steps.

1) PCA is used to unify the dimensions of the source domain and target domain data, and N samples are selected from each category to construct two-domain meta-training sets.
2) The spatial location information of labeled samples in the target domain is randomly reset to corresponding information of the first-order neighborhood, and samples with diverse spatial information are obtained to augment the sample space of the target-domain meta-training set.
3) First, NSIR is used to augment training samples, two-domain meta-training sets are input into the aggregation function. Then, RFSA and SIS are used to learn the importance of spectral band and spatial importance weight. Next, based on the learned weights, the discriminative spatial–spectral features are obtained via the aggregation function. Finally, the distance between the support set and query set of two domains in the spatial-spectral embedding space is calculated, and FSL is performed.
4) The distribution alignment is achieved through the adversarial process of the conditional domain discriminator and feature extractor.

The testing phase of GMTN is shown in Fig. 2. We use a feature extractor to learn target-domain embedding features with low intracategory dissimilarity and high intercategory dissimilarity, and exploit KNN to classify unlabeled target-domain samples.

C. Pseudocode of GMTN

The pseudocode of GMTN is given in Algorithm 1.

D. Data Augmentation Based on NSIR

On the one hand, the source of episodic training and the target of episodic training need to be similar. On the other hand, it is difficult to achieve domain adaptation due to limited samples. Therefore, we consider using data augmentation technology to expand the target-domain training set. Commonly

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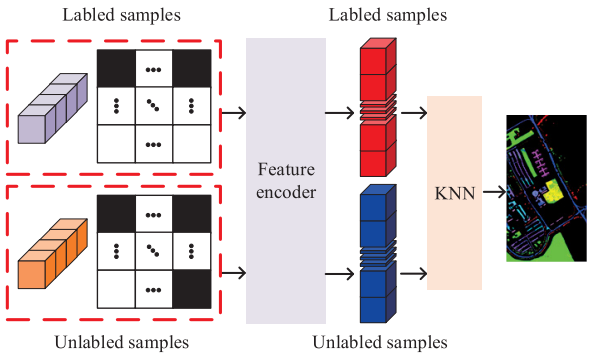

Fig. 2. Testing phase of GMTN.

used data augmentation methods for HSI classification are usually divided into three categories: spectral feature-based [33], image change-based [34], and noise-based [32]. The first method aims to generate new samples from each original sample based on the correlation between bands. The second method aims to obtain new samples by performing many transformations on HSI. The last one attempts to obtain new samples by adding noise to the original data. However, the above methods ignore the rich spatial information of HSIs, which affects the quality of newly generated samples and limits the improvement of classification performance. Therefore, the data augmentation method NSIR is proposed.

The working mechanism of NSIR is shown in the green-framed diagram in Fig. 1. First, randomly select a node from the first-order neighborhood of the target node. Then, the spatial information of the target node is reset to the spatial information of its first-order neighborhood node. Finally, every first-order neighborhood node is taken as the target node in turn, and the previous operation of spatial information reset is repeated to obtain abundant augmentation samples.

E. Feature Aggregation Based on GraphSAGE

The traditional GCN belongs to transductive learning, which makes it difficult to effectively characterize nodes unseen in training. Therefore, GCN is not suitable for cross-graph node representation. GraphSAGE can use node attribute information to summarize aggregation functions. Therefore, we embed it in the meta transfer framework to summarize the meta-knowledge that can be transferred in heterogeneous cross-domain scenarios.

According to [13], we train a set of aggregation functions that learn to aggregate feature information from the neighborhood nodes (pixels) to provide adequate characterization of samples in combination with the spectral and spatial information.

Define the graph as $G = (V, \varepsilon)$, the input feature as $\boldsymbol{x}_v$, $\forall v \in V$, and K as the depth parameter, $k \in \{1, \ldots, K\}$ aggregated local neighborhood feature information $h^k_{N_v}$ can be expressed as

$$h^k_{N_v} = \phi\left(\left\{h^{k-1}_u, \forall u \in N_v\right\}\right). \tag{1}$$

When $k = 0$, we have $h^0_v = \boldsymbol{x}_v$. N_v represents the neighborhood nodes (pixels) of v, and $\phi(\cdot)$ is the aggregation function, which is mean function here.

Algorithm 1 Training Process for GMTN

RANDOMSAMPLE (J, R) denotes a set of J elements chosen Uniformly at Random From Set R

Input: Raw hyperspectral data $\boldsymbol{X}^{\mathrm{s}}_{\mathrm{o}}$, $\boldsymbol{Y}^{\mathrm{s}}_{\mathrm{o}}$, $\boldsymbol{X}^{\mathrm{t}}_{\mathrm{o}}$, $\boldsymbol{Y}^{\mathrm{t}}_{\mathrm{o}}$
Output: The loss for a training episode
*/ Unify hyperspectral data dimensions as b in two domains
$\boldsymbol{X}^{\mathrm{s}} = \mathrm{PCA}(\boldsymbol{X}^{\mathrm{s}}_{\mathrm{o}}, b)$; $\boldsymbol{X}^{\mathrm{t}} = \mathrm{PCA}(\boldsymbol{X}^{\mathrm{t}}_{\mathrm{o}}, b)$
*/ Construct the meta-training set T
$T^{\mathrm{s}} = \{z^{\mathrm{s}}_1, \ldots, z^{\mathrm{s}}_k, \ldots, z^{\mathrm{s}}_{K^{\mathrm{s}}}\}$
$z^{\mathrm{s}}_k = \{(\boldsymbol{x}^{\mathrm{s}}_{k_1}, \boldsymbol{y}^{\mathrm{s}}_k), \ldots, (\boldsymbol{x}^{\mathrm{s}}_{k_i}, \boldsymbol{y}^{\mathrm{s}}_k) \ldots, (\boldsymbol{x}^{\mathrm{s}}_{k_N}, \boldsymbol{y}^{\mathrm{s}}_k)\}$
where $\boldsymbol{x}^{\mathrm{s}}_{k_i}$ is the i-th sample of category k in the source domain
$T^{\mathrm{t}} = \{z^{\mathrm{t}}_1, \ldots, z^{\mathrm{t}}_k, \ldots, z^{\mathrm{t}}_{K^{\mathrm{t}}}\}$
$z^{\mathrm{t}}_k = \{(\boldsymbol{x}^{\mathrm{t}}_{k_1}, \boldsymbol{y}^{\mathrm{t}}_k), \ldots, (\boldsymbol{x}^{\mathrm{t}}_{k_i}, \boldsymbol{y}^{\mathrm{t}}_k) \ldots, (\boldsymbol{x}^{\mathrm{t}}_{k_{l_{\mathrm{t}}}}, \boldsymbol{y}^{\mathrm{t}}_k)\}$
where l_{t} is the number of labeled samples of each category in the target domain
$T^{\mathrm{t}'} \leftarrow$Data augmentation for target domain hyperspectral data through NSIR
$T^{\mathrm{t}'} = \{z^{\mathrm{t}}_1, \ldots, z^{\mathrm{t}}_k, \ldots, z^{\mathrm{t}}_{K^{\mathrm{t}}}\}$
$z^{\mathrm{t}}_k = \{(\boldsymbol{x}^{\mathrm{t}}_{k_1}, \boldsymbol{y}^{\mathrm{t}}_k), \ldots, (\boldsymbol{x}^{\mathrm{t}}_{k_i}, \boldsymbol{y}^{\mathrm{t}}_k) \ldots, (\boldsymbol{x}^{\mathrm{t}}_{k_N}, \boldsymbol{y}^{\mathrm{t}}_k)\}$
*/ Meta-training
for i *in* 1 : episodes **do**
  */ Randomly select samples of C categories to form D from K categories of meta-training set
  $D \leftarrow$ RANDOMSAMPLE$(\{1, \ldots, K\}, C)$
  */ Randomly select support set S and query set Q to construct meta task and calculate category prototype c_k
  for k *in* $\{1, \ldots, C\}$ **do**
    $S_k \leftarrow$RANDOMSAMPLE(T_{D_k}, M)
    $Q_k \leftarrow$RANDOMSAMPLE$(T_{D_k}/S_k, E)$
    $c_k \leftarrow \frac{1}{M}\sum_{\boldsymbol{x}_i \in S_k} f_\Phi(\boldsymbol{x}_i)$
  end
  */ Calculate the loss of few-shot learning
  $L_{\mathrm{fsl}} = 0$
  for k *in* $\{1, \ldots, C\}$ **do**
    for $\boldsymbol{x}^{\mathrm{Q}}$ *in* Q_k **do**
      $$L_{\mathrm{fsl}} = L_{\mathrm{fsl}} + \frac{1}{CE} d\left(f_\Phi\left(\boldsymbol{x}^{\mathrm{Q}}\right), c_k\right) + \frac{1}{CE}\log\sum_{c=1}^{C}\exp\left(-d\left(f_\Phi\left(\boldsymbol{x}^{\mathrm{Q}}\right), c_k\right)\right)$$
    end
  end
  */ Calculate the loss of domain adaptation
  $F \leftarrow$Extract feature of samples
  $G \leftarrow$Predict categories of samples
  $L_{\mathrm{d}} \leftarrow$Calculate domain adaptation loss *by* Eq.(12)
  */ Calculate the total loss
  $L_{\mathrm{total}} = L_{\mathrm{fsl}} + \eta L_{\mathrm{d}}$
end

$h^k_{N_v}$ is then connected to h^{k-1}_v, and h^k_v can be expressed as

$$h^k_v = \sigma\left(\boldsymbol{W}^k \cdot \mathrm{CONCAT}\left(h^{k-1}_v, h^k_{N_v}\right)\right) \tag{2}$$

where CONCAT is a connected operation, $\boldsymbol{W}^k$ is learnable aggregation function and $\sigma(\cdot)$ is a nonlinear activation function using Tanh. Finally, the set of feature embeddings of resulting nodes (pixels) is $\boldsymbol{Z} = \{h^K_v, v \in V\}$.

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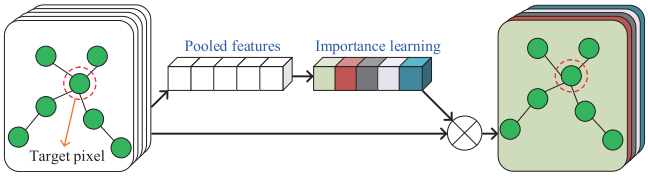

Fig. 3. Diagram of RFSA.

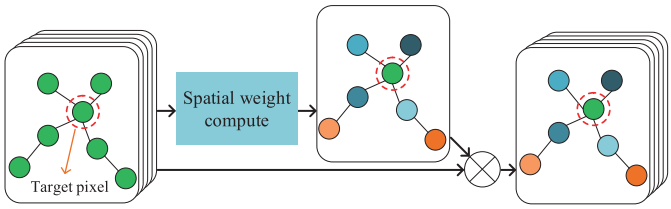

Fig. 4. Diagram of SIS.

F. Spectral Band Importance Learning Based on RFSA

Affected by illumination and weather, the collected HSI data is prone to contain noise [35]. Such phenomenon makes it difficult to extract discriminative features with limited training samples. The common solution is as follows [36], [37], [38]. First, global average pooling is conducted on the pixel blocks containing target pixel and its neighborhood pixels to acquire pooled features. Then, the pooled features are used to learn the importance of spectral bands and thus help the model extract spectral features. However, the global average pooling makes each pixel equally important to the pooled features, which leads to the interference of different-label neighborhood pixels (neighborhood pixels with different labels from the target pixel) to spectral band importance learning.

Therefore, RFSA is proposed to change the acquisition of pooled features in the above solution. Fig. 3 is the diagram of RFSA. Global average pooling is conducted on the neighborhood pixels

$$\boldsymbol{x}_{N_v} = \mathrm{AVE}(\{\boldsymbol{x}_u, \forall u \in N_v\}) \tag{3}$$

where $\mathrm{AVE}(\cdot)$ is global average pooling, N_v is the set of neighborhood pixels, and then the target pixel features are concatenated with $\boldsymbol{x}_{N_v}$ to obtain pooled features $\boldsymbol{G}$

$$\boldsymbol{G} = \mathrm{CONCAT}\left(\boldsymbol{x}_v, \boldsymbol{x}_{N_v}\right). \tag{4}$$

Thus, the importance of the target pixel features in the pooled features is increased, so that the interference of the different-label neighborhood pixels to spectral band importance learning is reduced. In this way, RFSA can achieve more effective spectral band importance learning, and support the discriminative features extraction of GMTN from the perspective of spectral information. The importance of spectral band weight $\boldsymbol{W}_I$ can be expressed by

$$\boldsymbol{W}_I = \sigma\left(\boldsymbol{W}_A \boldsymbol{G}\right) \tag{5}$$

where $\sigma(\cdot)$ is the activation function, $\boldsymbol{W}_A$ is the network parameter of RFSA. $\boldsymbol{W}_I$ is used to optimize the input features of GraphSAGE, thereby improving the discriminative features extraction of GMTN from the perspective of spectral information.

G. Weighting Neighborhood Pixels Based on SIS

Different neighborhood pixels make different contributions to the aggregated features. In the case of the few-shot classification scenario with high requirement on the discrimination of target pixel features, useful information should be enhanced while useless information should be suppressed during feature aggregation.

To this end, the SIS is proposed. Fig. 4 is the diagram of SIS. In view of the fact that the spectral features of different-label neighborhood pixels are usually quite different from those of target pixel, the spatial importance weights corresponding to neighborhood pixels are given based on the spectral feature similarity, so as to suppress information interference of different-label neighborhood pixels. The similarity between pixels can be calculated by

$$\omega_i^k = e^{\frac{-\left\|\boldsymbol{x}_v - \boldsymbol{n}_i^k\right\|^2}{\sigma}} \tag{6}$$

where $\boldsymbol{x}_v$ are the features of target pixel, $\boldsymbol{n}_i^k$ are the features of k-order neighborhood pixels.

Further considered, when two pixels have a large distance between them, different atmospheric and lighting conditions may make different objects have the same spectrum. As a result, different-label neighborhood pixels with similar spectral features to the target pixel will be wrongly assigned large weights. Therefore, the spatial importance weight is calculated based on the spectral feature similarity and distance constraint, which makes feature aggregation pay more attention to useful information, thus helping the GMTN extract discriminative spatial-spectral features. Spatial importance weight can be calculated by

$$\alpha_i^k = \lambda_k \omega_i^k \tag{7}$$

where λ_k is distance constraint, which can be calculated by

$$\lambda_k = \frac{1}{1 + e^{\frac{-1}{D_k}}} \tag{8}$$

where D_k is the spatial coefficient, which can be calculated by

$$D_k = \frac{1}{U_k} \sum_{u=1}^{U_k} \left\|d_v - d_u^k\right\|_2 \tag{9}$$

where U_k is the number of k-order neighborhood pixels of the target pixel, d_v and d_u are the coordinates of the target pixel and its neighborhood pixels, respectively.

H. Source and Target Domain FSL

First, the aggregation functions are used to obtain the spatial-spectral aggregation features of two domains, and then the Euclidean distance between the support and query sets is measured to realize FSL. The process of FSL is shown in Fig. 5. We randomly select C categories from the meta-training

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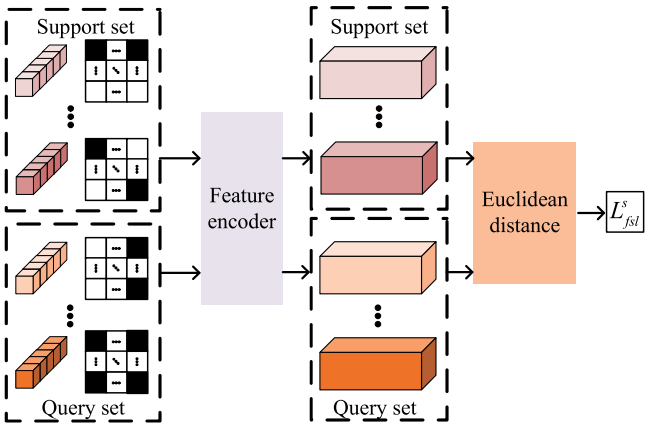

Fig. 5. Diagram of FSL.

set, M samples from each category to form a support set $S = \{(\boldsymbol{x}_i^S, \boldsymbol{y}_i^S)\}_{i=1}^{C\times M}$, and E samples to form a query set $Q = \{(\boldsymbol{x}_j^Q, \boldsymbol{y}_j^Q)\}_{j=1}^{C\times E}$. Here, the labels of the C categories are not the original labels but the meta-training labels reassigned in the order of category sample. This setting allows the aggregation function to learn a transferable aggregation function so that the aggregation features of different-category target pixels show large differences between categories. During the training phase, the category probability is used to calculate the loss, and then the network parameters are updated through back-propagation. The probability score of query set sample $\boldsymbol{x}_j^Q$ belonging to the cth category is

$$P\left(\boldsymbol{y}_j^Q{=}c|\boldsymbol{x}_j^Q \in Q\right){=}\frac{\exp\left(-d\left(f_\psi\left(\boldsymbol{x}_j^Q\right), \boldsymbol{r}_c\right)\right)}{\sum_{c=1}^{C}\exp\left(-d\left(f_\psi\left(\boldsymbol{x}_j^Q\right), \boldsymbol{r}_c\right)\right)} \quad (10)$$

where $d(\cdot)$ denotes the distance metric function, $\boldsymbol{r}_c$ is the sample of the support set of category c, $f_\psi(\cdot)$ is the feature extraction network, ψ is the network parameter, $\boldsymbol{x}_j^Q$ is the sample query set, and $\boldsymbol{y}_j^Q$ is the label of $\boldsymbol{x}_j^Q$. The loss of FSL is

$$L_{\text{fsl}} = E_{S,Q}\left[-\sum_{(x,y)\in Q}\log p_\psi(y=c|x)\right]. \quad (11)$$

I. Conditional Domain Adaptation

The data in the source and target domains are heterogeneous, which makes it difficult to align the distributions under few-shot scenarios. The conditional domain adversarial strategy [39], which can fully capture the multimodal structures behind complex data distributions, aligns the two-domain distributions through an adversarial process of conditional domain discriminator and feature extractor. The optimization goal of the conditional domain discriminator is

$$\begin{aligned}\min_D \max_{F,G}\ L_d = &-E_{x_i^s\sim P_s(x)}\log\left[D\left(T\left(H_i^s\right)\right)\right]\\ &-E_{x_i^t\sim P_t(x)}\log\left[D\left(T\left(H_i^t\right)\right)\right] \quad (12)\end{aligned}$$

where $\boldsymbol{x}_i^s$ and $\boldsymbol{x}_j^t$ are the source-domain sample and the target-domain sample, respectively. $P_s(x)$ and $P_t(x)$ are the source distribution and target distribution, respectively. D is the conditional domain discriminator, F_i^s and F_j^t are embedded features of the source and target domains, respectively, G_i^s and G_j^t are category prediction vectors for both domains. T is the mapping operation, which can be expressed as

$$T(H) = T_\otimes(F, H) \quad (13)$$

where $\otimes$ is the multilinear map.

J. Total Loss

The total loss of both the source and target domains can be expressed as

$$L_{\text{total}} = L_{\text{fsl}} + \eta L_d \quad (14)$$

where η is the trade-off factor.

III. Experiment

A. HSI Datasets

To demonstrate the superiority of our method, four HSI datasets were selected, including Chikusei, Salinas (SA), Pavia University (PU), and Indian Pines (IP). We choose Chikusei as the source domain and SA, PU, and IP as the target domains.

The Chikusei dataset is an HSI acquired by researchers using HVNIR-C sensors in Chikusei, Japan. It contains 2517 × 2335 pixels, 128 bands, and 19 categories.

The SA dataset is an HSI collected from the Salinas Valley, USA. It contains 512 × 217 pixels, 204 bands, and 16 categories.

The PU dataset is an HSI acquired from the university of Pavia. It contains 610 × 340 pixels, 103 bands, and nine categories.

The IP dataset is an HSI collected from Northwest Indiana. It contains 145 × 145 pixels, 200 bands, and 16 categories. The specific surface object names and pixels of four HSI datasets can be found in Table I.

B. Experimental Setting

To verify the effectiveness of GMTN, SVM [4], DCCN [10], GraphSAGE [13], deep FSL (DFSL) [29], siamese-prototype network (SPnet) [26], and deep cross-domain FSL (DCFSL) [32] are chosen as the comparison methods.

1) *Traditional Machine Learning Method:* SVM.
2) *Deep Learning Method:* DCCN.
3) *Graph Learning Method:* GraphSAGE.
4) *Meta-Learning Methods:* SPnet and DFSL.
5) *Meta Transfer Method:* DCFSL.

To ensure the fairness of comparison, we randomly selected five samples from each category in the target domain to participate in training. For some meta-learning methods and meta-transfer methods, we randomly select 200 labeled samples from each category in the source domain, and randomly select five labeled samples from each category in the target domain. Each episode is a C-way, Z-shot classification task, C is set to the number of the target-domain categories, Z is set to 1, that is, only one labeled sample is selected as the support

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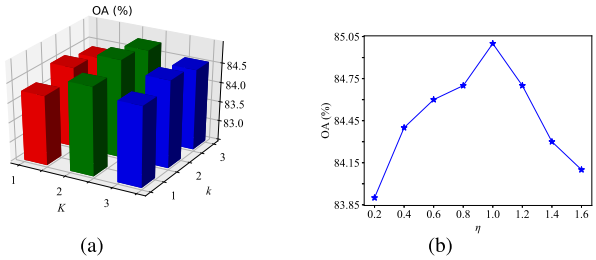
TABLE I
DESCRIPTION OF HSI DATASETS

	Chikusei			PU			SA			IP	
	Surface object	Pixels		Surface object	Pixels		Surface object	Pixels		Surface object	Pixels
C1	Water	2845	P1	Asphalt	6631	S1	Brocoli_green_weeds_1	2009	I1	Alfalfa	46
C2	Bare soil(school)	2859	P2	Meadows	18649	S2	Brocoli_green_weeds_2	3726	I2	Corn-notill	1428
C3	Bare soil(park)	286	P3	Gravel	2099	S3	Fallow	1976	I3	Corn-mintil	830
C4	Bare soil(farmland)	48525	P4	Trees	3064	S4	Fallow_rough_plow	1394	I4	Corn	237
C5	Natural plants	4297	P5	Sheets	1345	S5	Fallow_smooth	2678	I5	Grass-pasture	483
C6	Weeds in farmland	1108	P6	Bare soil	5029	S6	Stubble	3959	I6	Grass-trees	730
C7	Forest	20516	P7	Bitumen	1330	S7	Celery	3579	I7	Grass-pasture-mowed	28
C8	Grass	6515	P8	Bricks	3682	S8	Grapes_untrained	11271	I8	Hay-windrowed	478
C9	Rice field(grown)	13369	P9	Shadow	947	S9	Soil_vinyard_develop	6203	I9	Oats	20
C10	Rice field(first stage)	1268				S10	Corn_senesced_green_weeds	3278	I10	Soybean-notill	972
C11	Row crops	5961				S11	Lettuce_romaine_4wk	1068	I11	Soybean-mintill	2455
C12	Plastic house	2193				S12	Lettuce_romaine_5wk	1927	I12	Soybean-clean	593
C13	Manmade(non-dark)	1220				S13	Lettuce_romaine_6wk	916	I13	Wheat	205
C14	Manmade(dark)	7664				S14	Lettuce_romaine_7wk	1070	I14	Woods	1265
C15	Manmade(black)	431				S15	Vinyard_untrained	7268	I15	Buildings-Grass-Trees-Drives	386
C16	Manmade(red)	222				S16	Vinyard_vertical_trellis	1807	I16	Stone-Steel-Towers	93
C17	Manmade grass	4040									
C18	Asphalt	801									
C19	Paved ground	145									
	Total	77592		Total	42776		Total	54129		Total	10249



(a) (b)

Fig. 6. OAs achieved by GMTN under different parameter settings (PU). (a) OA versus K and k. (b) OA versus η.

set. E is the number of samples in query set, which is set to 19 according to [32].

The results are the average of ten experiments. Four evaluating indexes are selected for the experiment, including overall accuracy (OA), accuracy of each category, Kappa coefficient, and standard deviation.

C. Parameter Setting and Analysis

We analyze the influence of different parameter settings on the performance of GMTN, including the depth parameter K of GraphSAGE, the depth parameter k of NSIR, and the trade-off factor η. For simplicity, other parameters are settled as the same, i.e., the number of maximal episodes is set at 1000, the optimizer is RMSProp, and the learning rate is 0.002.

The parameter settings were analyzed using PU as an example. First, we study the effects of K, k, and η on the OA. Fig. 6 shows the following observations. First, with the increase of K, OA shows a trend of increasing first and then decreasing. When $K = 1$, the number of aggregated neighborhood pixels is limited, and the spatial information of HSI is difficult to be fully utilized. When $K = 3$, the neighborhood pixels far from the target pixel are aggregated, and different categories of aggregated pixels lead to information interference. When $K = 2$, OA reaches maximum.

TABLE II
CLASSIFICATION PERFORMANCE ON PU (FIVE LABELED SAMPLES EACH CATEGORY)

Category	SVM	DCCN	GraphSAGE	DFSL	SPnet	DCFSL	GMTN
P1	62.75	71.55	47.83	60.67	81.80	**82.20**	76.99
P2	57.50	66.24	66.86	93.91	82.53	87.74	**96.65**
P3	55.06	**77.45**	62.03	69.52	62.61	67.46	55.16
P4	85.45	93.03	85.13	76.66	87.32	**93.16**	70.32
P5	98.96	99.93	96.94	98.28	**100**	99.49	98.88
P6	57.66	**90.80**	64.23	37.68	67.95	77.32	76.89
P7	**94.34**	88.64	85.59	90.26	82.11	81.18	78.26
P8	56.87	77.70	60.70	58.20	**78.73**	66.73	74.15
P9	99.89	**100**	99.58	98.62	99.04	98.66	98.73
OA (%)	63.54 ±4.15	75.89 ±1.32	66.39 ±3.98	76.77 ±1.32	80.64 ±1.21	83.65 ±1.86	**84.97 ±1.95**
Kappa (%)	54.72 ±4.36	70.02 ±1.17	57.96 ±4.23	68.49 ±1.53	74.74 ±1.35	78.70 ±2.15	**79.77 ±2.18**

Second, with the increase of k, OA tends to rise first and then drop down. When $k = 1$, only the first-order neighborhood information is used for data augmentation, leading to limited diversity of expanded samples. When $k = 3$, the influence of information interference is further expanded due to the data augmentation of different categories of aggregated pixels. When $k = 2$, OA reaches the highest. Third, with the increase of η, OA first increases and then decreases. When η is small, the two domains cannot be fully adapted. In contrast, it may lead to over-adaptation.

Note that for simplicity, the parameter settings for the other datasets are consistent with PU, i.e., we set $K = 2$, $k = 2$, and $\eta = 1$.

D. Comparison Results Analysis

As can be seen from Tables II–IV and Figs. 7–10, the following results obtained.

1) Compared with traditional classification methods, the deep learning method and graph learning method achieve better classification performance, because the deep learning method can extract high-level features,

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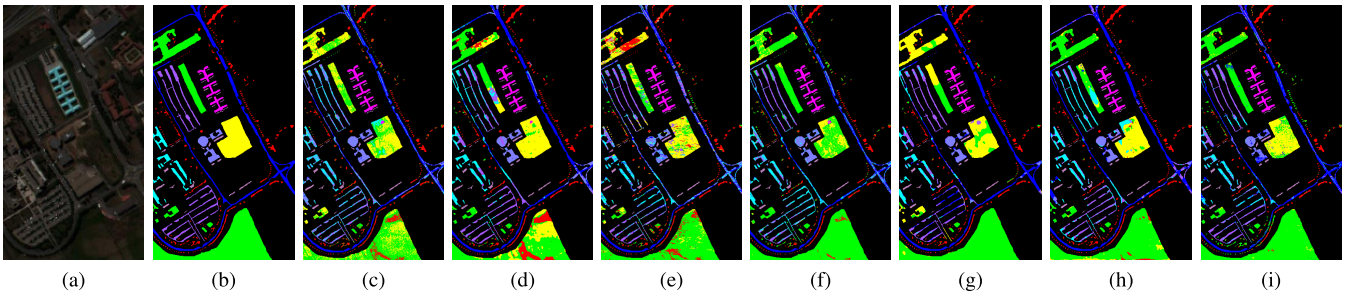

Fig. 7. Classification maps on PU. (a) False-color image. (b) Label map. (c) SVM. (d) DCCN. (e) GraphSAGE. (f) DFSL. (g) SPnet. (h) DCFSL. (i) GMTN.

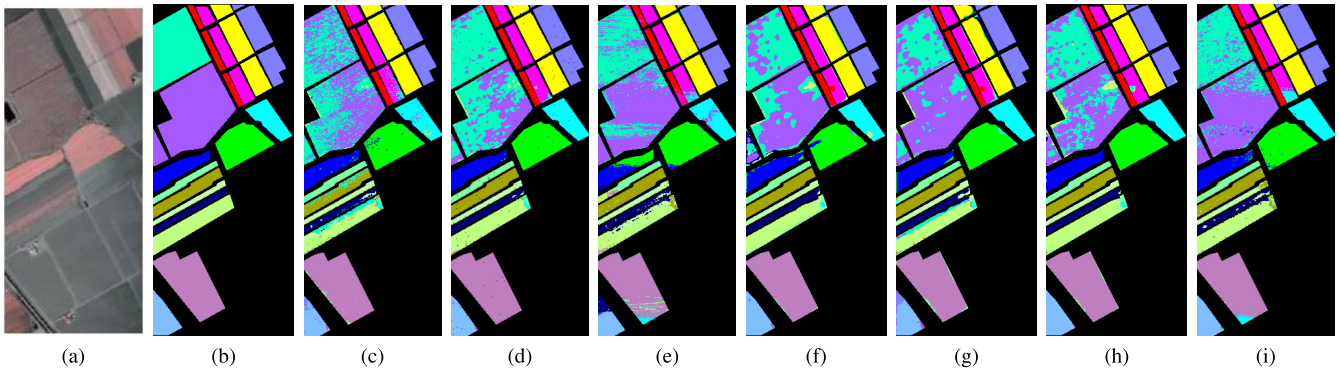

Fig. 8. Classification maps on SA. (a) False-color image. (b) Label map. (c) SVM. (d) DCCN. (e) GraphSAGE. (f) DFSL. (g) SPnet. (h) DCFSL. (i) GMTN.

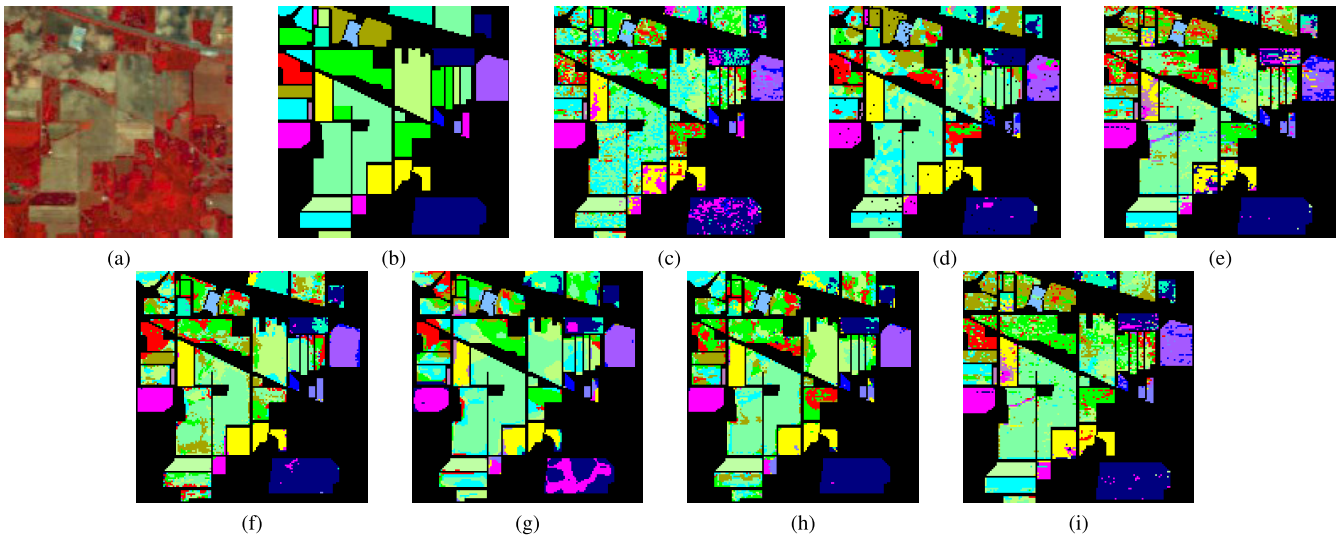

Fig. 9. Classification maps on IP. (a) False-color image. (b) Label map. (c) SVM. (d) DCCN. (e) GraphSAGE. (f) DFSL. (g) SPnet. (h) DCFSL. (i) GMTN.

while the graph learning method can use spatial information.

2) Since the transferable meta-knowledge can be learned from the source domain, SPnet and DFSL obtain more competitive results than the classification methods only using the target-domain knowledge.
3) The meta transfer method DCFSL achieves the second classification accuracy, which is due to the fact that the meta transfer method can learn the transferable meta-knowledge in the source domain on the one hand. On the other hand, the domain adaptation of the source and target domains is extended, which aligns the two-domain distributions.
4) As can be seen from Figs. 7–9, compared with GraphSAGE, which also extracts spatial-spectral features by feature aggregation, GMTN has fewer misclassification at the edge of category. This is because SIS can reduce the interference of different-label neighborhood pixels.
5) From Fig. 10, the features learned by GMTN show higher similarity of intracategory than those by DCFSL, which is also a meta transfer method, and the edges of intercategory features are clearer.

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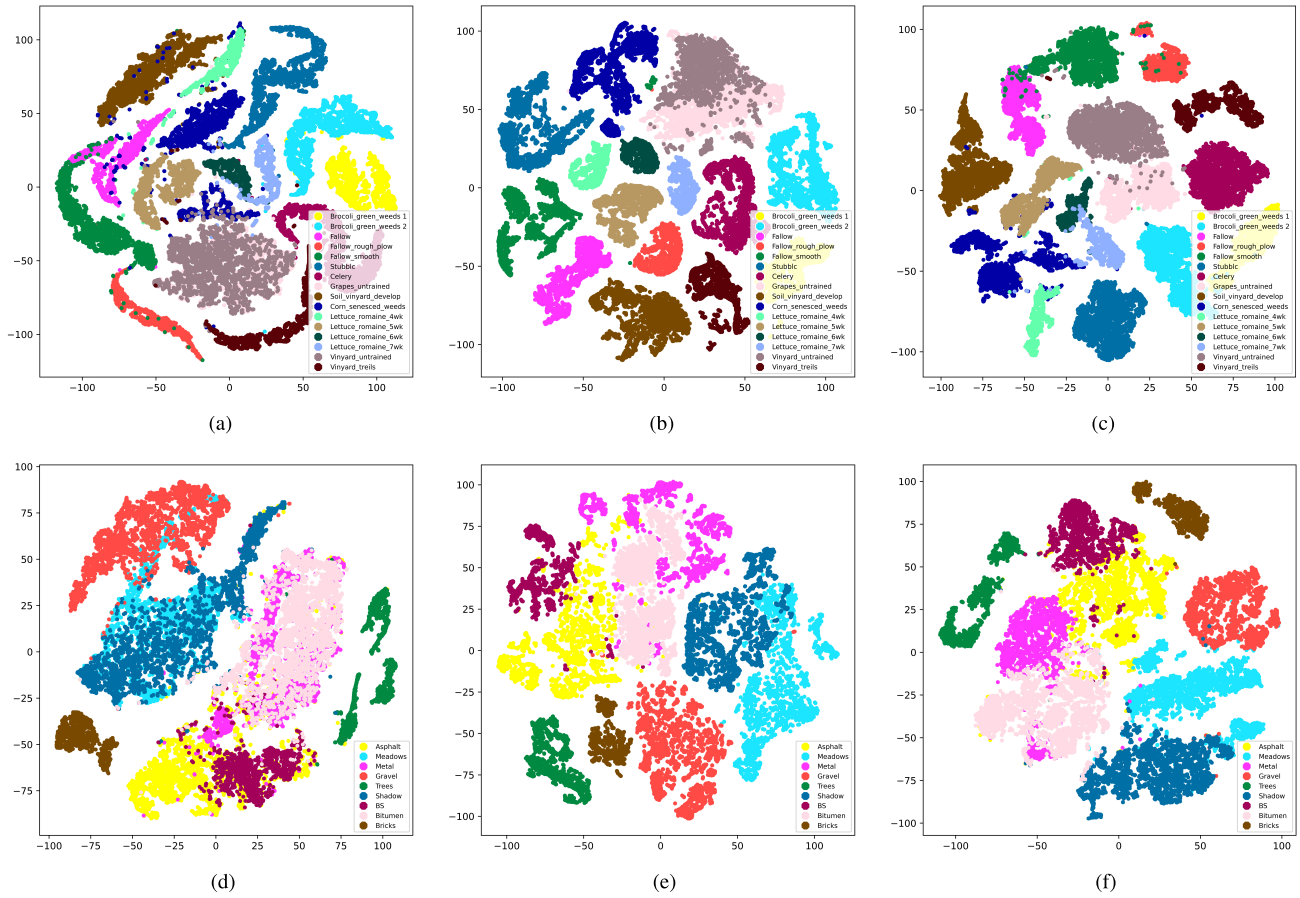

Fig. 10. T-distributed stochastic neighborhood embedding of HSI features. (a) Original SA features. (b) Features learned by DCFSL on SA. (c) Features learned by GMTN on SA. (d) Original PU features. (e) Features learned by DCFSL on PU. (f) Features learned by GMTN on PU.

TABLE III
CLASSIFICATION PERFORMANCE ON SA (FIVE LABELED SAMPLES EACH CATEGORY)

Category	SVM	DCCN	GraphSAGE	DFSL	SPnet	DCFSL	GMTN
S1	98.35	**99.95**	59.53	86.27	96.11	99.40	97.00
S2	97.15	**100**	97.37	96.72	96.69	99.76	98.20
S3	92.39	97.25	94.01	96.50	99.39	91.96	**99.49**
S4	96.98	**99.64**	84.45	97.62	99.50	99.54	98.92
S5	89.60	**96.59**	91.17	93.09	90.98	92.70	86.00
S6	97.42	99.82	98.69	**100**	85.03	99.52	99.32
S7	**99.47**	**99.47**	98.27	99.44	99.25	98.88	99.24
S8	51.48	47.74	79.54	70.89	84.59	74.57	**84.94**
S9	98.19	**100**	92.90	97.95	97.03	99.59	96.72
S10	60.53	**91.07**	68.84	66.54	78.70	86.42	89.26
S11	89.37	94.42	69.05	82.03	95.96	**96.61**	93.60
S12	81.53	**100**	98.07	96.46	99.12	99.93	96.72
S13	98.90	98.79	96.82	98.68	**99.56**	99.30	90.11
S14	87.70	97.17	89.39	89.95	91.55	**98.85**	86.37
S15	59.62	**86.44**	77.34	82.99	64.70	75.38	83.31
S16	84.74	91.82	74.64	74.08	83.19	92.22	**98.83**
OA (%)	78.88 ±2.86	85.95 ±1.29	85.26 ±2.16	86.52 ±0.96	87.57 ±0.89	89.34 ±1.23	**91.80 ±1.06**
Kappa (%)	76.58 ±3.12	84.47 ±1.04	83.62 ±2.32	85.05 ±1.25	86.14 ±1.26	88.17 ±1.34	**90.88 ±1.15**

TABLE IV
CLASSIFICATION PERFORMANCE ON IP (FIVE LABELED SAMPLES EACH CATEGORY)

Category	SVM	DCCN	GraphSAGE	DFSL	SPnet	DCFSL	GMTN
I1	85.36	**99.14**	68.27	72.22	92.68	95.12	56.10
I2	36.19	21.86	36.43	44.43	36.47	41.95	**53.76**
I3	45.33	48.42	44.71	**55.98**	43.52	50.19	48.24
I4	46.98	72.25	59.72	73.57	62.93	**87.07**	79.31
I5	83.47	**87.74**	81.92	87.53	79.71	86.40	78.45
I6	74.76	**98.89**	69.82	89.72	82.48	85.51	77.93
I7	91.30	**100**	89.92	77.78	**100**	**100**	**100**
I8	66.17	66.45	58.25	72.22	**97.25**	93.23	84.14
I9	86.67	**100**	89.80	**100**	**100**	**100**	93.33
I10	32.99	26.61	30.58	45.11	35.47	**55.84**	31.85
I11	40.53	57.59	63.98	49.98	71.39	68.29	**75.95**
I12	19.73	47.17	51.28	37.74	34.18	30.00	**64.96**
I13	97.50	**100**	97.70	94.87	96.00	**100**	99.00
I14	64.60	84.38	75.37	77.29	85.40	89.37	**89.44**
I15	49.60	**71.81**	61.53	49.73	63.26	58.53	65.88
I16	82.96	91.57	86.22	96.39	**100**	97.72	86.36
OA (%)	49.37 ±4.63	58.91 ±1.64	55.35 ±3.98	59.49 ±1.21	63.22 ±1.15	66.77 ±1.85	**68.36 ±1.64**
Kappa (%)	43.34 ±4.72	53.67 ±1.45	52.47 ±4.05	54.49 ±1.31	58.10 ±1.28	62.64 ±1.95	**64.03 ±1.72**

E. Analysis of the Influence of Different Training Samples Proportions

To show the generalization ability of GMTN, we randomly select a certain proportion of labeled samples in the source domain and target domains as training samples, and remaining samples of the target domains are reserved as the testing set. Table V shows the OAs obtained by different methods on the

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TABLE V
CLASSIFICATION PERFORMANCE (OA, %) WITH DIFFERENT PROPORTIONS OF SOURCE-DOMAIN TRAINING SAMPLES

Dataset	Proportions of training samples		DFSL	SPnet	DCFSL	GMTN
	Source domain	Target domain				
PU	4.64%	0.11%	76.77±1.32	80.64±1.21	83.65±1.86	**84.97±1.95**
	3.71%	0.11%	75.69±1.23	79.32±1.13	81.47±1.64	**82.84±1.86**
	2.78%	0.11%	74.93±1.42	78.36±1.35	80.69±1.67	**81.95±1.88**
	1.86%	0.11%	73.65±1.26	78.12±1.13	80.15±1.42	**81.23±1.83**
	0.93%	0.11%	71.26±1.31	71.17±1.25	77.31±1.36	**79.42±1.87**
SA	4.64%	0.15%	86.52±1.02	87.57±0.89	89.34±1.23	**91.80±1.06**
	3.71%	0.15%	84.98±1.12	86.13±1.08	88.64±1.28	**90.72±1.36**
	2.78%	0.15%	84.12±1.27	85.65±1.12	87.96±1.07	**89.97±1.25**
	1.86%	0.15%	83.26±1.26	84.34±1.17	86.26±1.35	**88.93±1.01**
	0.93%	0.15%	81.45±1.33	82.15±1.25	84.65±1.18	**86.17±0.89**
IP	4.64%	0.78%	59.49±1.21	63.22±1.15	66.77±1.85	**68.36±1.64**
	3.71%	0.78%	58.32±1.42	62.54±0.89	65.85±1.47	**67.62±1.77**
	2.78%	0.78%	57.68±1.07	61.87±1.86	65.14±1.43	**67.05±1.57**
	1.86%	0.78%	56.12±1.26	60.45±1.06	63.84±1.37	**65.57±1.48**
	0.93%	0.78%	54.38±1.35	58.14±1.12	62.34±1.43	**63.27±1.68**

TABLE VI
CLASSIFICATION PERFORMANCE (OA, %) WITH DIFFERENT PROPORTIONS OF TARGET-DOMAIN TRAINING SAMPLES

Dataset	Proportions of training samples		DFSL	SPnet	DCFSL	GMTN
	Source domain	Target domain				
PU	4.64%	0.11%	76.77±1.23	80.64±1.32	83.65±1.92	**84.97±1.82**
	4.64%	0.08%	70.28±1.14	78.26±1.53	80.36±1.71	**81.15±1.21**
	4.64%	0.06%	68.18±1.25	73.25±1.43	74.12±1.53	**76.35±1.13**
	4.64%	0.04%	64.72±1.14	66.17±1.21	68.25±1.41	**70.21±1.32**
	4.64%	0.02%	52.28±1.26	55.38±1.17	57.65±1.37	**60.59±1.47**
SA	4.64%	0.15%	86.52±1.41	87.57±0.93	89.34±1.44	**91.80±1.13**
	4.64%	0.12%	85.38±1.28	86.29±1.16	88.69±1.36	**89.45±1.33**
	4.64%	0.09%	82.34±1.29	85.12±1.25	85.63±1.24	**86.23±1.15**
	4.64%	0.06%	76.35±1.28	79.12±1.24	80.92±1.34	**83.12±1.21**
	4.64%	0.03%	68.54±1.41	70.26±1.21	72.71±1.26	**77.49±0.86**
IP	4.64%	0.78%	59.49±1.35	63.22±1.25	66.77±1.73	**68.36±1.31**
	4.64%	0.63%	58.29±1.39	61.84±0.77	62.89±1.54	**65.16±1.62**
	4.64%	0.47%	55.48±1.03	58.28±1.72	58.52±1.37	**61.32±1.42**
	4.64%	0.31%	48.25±1.31	55.36±1.16	57.63±1.42	**59.24±1.36**
	4.64%	0.16%	40.13±1.17	41.15±1.15	42.75±1.47	**46.81±1.47**

three datasets by fixing the proportions of target-domain training samples and changing the proportion of source-domain training samples from 0.93% (40 samples per category) to 4.64% (200 samples per category). Table VI shows the OAs obtained by different methods on the three datasets by fixing the proportion of source-domain training samples and changing the proportions of target-domain training samples from 0.02%, 0.03%, and 0.16% (one sample per category) to 0.11%, 0.15%, and 0.78% (five samples per category), respectively. In Tables V and VI, the remaining samples of the target-domain datasets (PU, SA, and IP) are reserved as the testing set.

As can be seen from Tables V and VI, with the increase in the proportion of training samples, the OA of GMTN is improved continuously, and when the proportion of training samples in the source domain and target domain is only 3.71% and 0.15%, respectively, GMTN can obtain an OA of more than 90% on SA. In addition, compared with the state-of-the-art few-shot HSI classification methods such as SPnet and DCFSL, GMTN can always obtain the best classification results on all datasets with different proportions of training samples. As can be seen from Tables II–IV, for cross-domain few-shot classification, the OAs of the proposed method on PU, SA, and IP are 9.08%, 5.85%, and 9.45% higher than that of DCCN, respectively, which is the state-of-the-art deep learning method. All the results show that our proposed GMTN has strong adaptability to the change of the proportion of training samples and has good generalization ability.

TABLE VII
ABLATION EXPERIMENTS

Technique	GMN	GMTN-A	GMTN-B	GMTN-C	GMTN-D	GMTN
RFSA					✓	✓
SIS				✓		✓
NSIR			✓	✓	✓	✓
DA		✓	✓	✓	✓	✓

F. Ablation Study

To demonstrate the contribution of different techniques of GMTN to the ultimate classification result, we conducted ablation experiments on three datasets, and the results are shown in Tables VII and VIII.

It can be observed from Tables VII and VIII that the following holds.

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TABLE VIII
IMPACT OF DIFFERENT TECHNIQUES ON OA (%)

Dataset	GMN	GMTN-A	GMTN-B	GMTN-C	GMTN-D	GMTN
PU	77.42	82.85	84.08	84.49	84.35	**84.97**
	±1.15	±1.26	±1.31	±1.24	±1.75	**±1.95**
SA	86.52	88.57	90.85	91.18	91.04	**91.80**
	±1.35	±1.17	±1.22	±1.22	±1.14	**±1.06**
IP	60.98	65.92	67.23	67.82	67.76	**68.36**
	±1.14	±1.08	±1.12	±1.18	±1.51	**±1.64**

TABLE IX
COMPUTATIONAL COST ON EACH DATASET WITH DIFFERENT METHODS

Dataset	Index	SVM	DCCN	GraphSAGE	DFSL	SPnet	DCFSL	GMTN
PU	Time (s)	2.96	291.28	289.63	756.32	617.29	1253.21	348.60
	FLOPs (M)	-	15.63	3.21	41.13	41.13	46.19	6.97
	Param (M)	-	3.42	1.19	3.49	3.49	42.59	7.84
SA	Time (s)	3.45	338.62	342.32	1032.34	715.14	1267.36	354.80
	FLOPs (M)	-	17.80	3.67	41.13	41.13	47.01	6.97
	Param (M)	-	3.61	1.23	3.49	3.49	42.69	7.84
IP	Time (s)	1.87	189.26	241.37	586.26	475.35	1250.28	318.60
	FLOPs (M)	-	17.40	3.67	41.13	41.13	46.98	6.97
	Param (M)	-	3.61	1.23	3.49	3.49	42.69	7.84

1) Compared with GMT, GMTN-A shows higher classification accuracy. This is because DA reduces the distribution difference between the two domains and realizes knowledge transfer.
2) Compared with GMTN-A, GMTN-B and GMTN-C show superior classification performance. The reasons are as follows. SIS assigns importance weights to neighborhood pixels, which can mitigate the information interference generated during feature aggregation; NSIR effectively expands the training set based on spatial information, so that the network can be fully trained.
3) Compared with GMTN-B, GMTN-D has better classification performance, since RFSA models the importance of spectral bands, which makes the network pay more attention to identifiable bands.

G. Computational Cost

In this section, we show the computational cost including computational time, floating point operations (FLOPs), and number of network parameters for different methods in Table IX.

It can be observed from Table IX that the following holds.

1) Compared with SVM, DCCN, and GraphSAGE, GMTN consumes more computational time because it requires extra computational cost to align the two-domain distributions and train the model using the source-domain samples.
2) Compared with DFSL, SPnet, and DCFSL, GMTN has less computational time, FLOPs, and parameters due to its simple structure.

According to the above analysis, the improvement in classification accuracy of GMTN requires a slight sacrifice of computational cost.

IV. CONCLUSION

To solve the heterogeneous few-shot HSI classification task, where the training and testing sets have different categories, GMTN is proposed. GMTN has the following advantages: 1) it can solve the problem of the difficult extraction of transferable meta-knowledge caused by heterogeneous few-shot scenarios; 2) it can not only endow the augmented samples with rich spatial information but also enable augmented samples to aggregate diverse adjacent spectral information; 3) it can fully suppress the band redundancy so that the network will pay more attention to the bands with discriminative features; and 4) it can alleviate the information interference caused by different categories when the information of neighborhood pixels is aggregated.

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